



QP CODE: 24805306



24805306

Reg No :

Name :

MCA DEGREE EXAMINATIONS, NOVEMBER 2024

Third Semester

Master of Computer Application

Core - MCACT301 - MACHINE LEARNING TECHNIQUES

2020 Admission Onwards

CABF04D6

Time: 3 Hours

Maximum: 75 Marks

Part A

*Answer any **ten** questions*

*Each question carries **3** marks*

1. Explain supervised learning with an example.
2. Write short notes on classification.
3. Give two examples of classification problem.
4. What is a tree pruning?
5. What is a margin in SVM?
6. What is the concept of Dimensionality Reduction?
7. Differentiate between Factor analysis and PCA?
8. Explain about Clustering.
9. Define Hierarchical clustering.
10. Write a note on boosting and bagging.
11. What is Artificial Intelligence?
12. Define a Perceptron?

(10×3=30 marks)





Part B

Answer **all** questions

Each question carries **9** marks

13. a) Compare different types of machine learning methods.

OR

b) Write short notes on i) Artificial Intelligence. ii) Machine Learning. iii) Deep Learning

14. a) Discuss about learning class in Machine Learning with an example

OR

b) Explain about different Univariate Trees with example.

15. a) Describe feature Extraction and Feature selection method in machine learning.

OR

b) Discuss the concept of principal Component Analysis.

16. a) What are the different types of similarity measures in clustering?

OR

b) Explain about the terms i) Density Based clustering ii) Hierarchical clustering

17. a) Explain about Perceptron neural network and its training algorithm.

OR

b) Explain about Backpropagation feedforward NN and its training algorithm.

(5×9=45 marks)

